WEEK 6 – PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

1. Introduction

The Product Review and Rating Management System is used to collect and manage customer feedback about purchased products. Customers can provide ratings and reviews after completing their orders. The system stores customer opinions and helps analyse product performance. It also helps identify highly rated products and products that require improvement.

2. Objective

The main objectives of this system are:

- To create review and rating tables.
- To store customer feedback and product ratings.
- To update and delete reviews when required.
- To retrieve product review details.
- To calculate average product ratings.
- To identify highly rated products.
- To generate product rating and customer feedback reports.

3. Create Review and Rating Tables

A review table is created to store customer feedback about purchased products. It contains details such as Review ID, Customer ID, Product ID, Rating, Review Comment, and Review Date. Primary key and foreign key constraints are used to maintain relationships with the existing Customer and Product tables.

A CHECK constraint is used to ensure that customer ratings are between 1 and 5.

```
SQL> CREATE TABLE Review(ReviewID VARCHAR2(7), CustomerID NUMBER(38), ProductID VARCHAR(7), Rating NUMBER(1), ReviewTest VARCHAR2(50), ReviewDate DA
TE, PRIMARY KEY(ReviewID), FOREIGN KEY(CustomerID) REFERENCES Customer(Customer_ID), FOREIGN KEY(ProductID) REFERENCES Product(Product_ID));

Table created.
```

```
SQL> desc Review;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 REVIEWID                                  NOT NULL VARCHAR2(7)
 CUSTOMERID                                         NUMBER(38)
 PRODUCTID                                          VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEWTEST                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

4. Store Customer Feedback and Ratings

Customer reviews and ratings are inserted into the database after a product is purchased. Ratings are stored on a scale of 1 to 5 stars. The review comments can be updated if customers want to modify their feedback. Inappropriate or invalid reviews can also be removed from the database.

Tasks:

1. Add reviews for purchased products.

```
SQL> INSERT INTO REVIEW VALUES ('REV0001', 101, 'P301', 5, 'Excellent backpack', DATE '2026-01-20');

1 row created.

SQL> INSERT INTO REVIEW VALUES ('REV0002', 102, 'P302', 4, 'Good football', DATE '2026-01-21');

1 row created.

SQL> INSERT INTO REVIEW VALUES ('REV0003', 103, 'P303', 5, 'Excellent car cover', DATE '2026-02-02');

1 row created.

SQL> INSERT INTO REVIEW VALUES ('REV0004', 104, 'P304', 3, 'Average pet food', DATE '2026-02-03');

1 row created.

SQL> INSERT INTO REVIEW VALUES ('REV0005', 105, 'P305', 5, 'Very good backpack', DATE '2026-02-15');

1 row created.
```

```
SQL> Select * from Review;

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0001        101 P301             5
Excellent backpack                                  20-JAN-26

REV0002        102 P302             4
Good football                                       21-JAN-26

REV0003        103 P303             5
Excellent car cover                                 02-FEB-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0004        104 P304             3
Average pet food                                    03-FEB-26

REV0005        105 P305             5
Very good backpack                                  15-FEB-26
```

2. Store customer ratings from 1 to 5 stars.

```
SQL> ALTER TABLE Review ADD CONSTRAINT CHK_Rate CHECK(Rating >= 1 AND Rating <= 5);

Table altered.
```

3. Update review comments if required.

```
SQL> UPDATE REVIEW SET REVIEWTEST = 'Good quality but could be improved' WHERE REVIEWID = 'REV0004';

1 row updated.
```

4. Remove inappropriate or invalid reviews.

```
SQL> DELETE FROM REVIEW WHERE REVIEWID = 'REV0005';

1 row deleted.
```

5. Retrieve Product Review Details

SQL queries are used to retrieve review information from the database. The system can display all reviews for a particular product, customer names with their reviews, products having maximum reviews, and recent customer feedback. It can also display reviews with ratings above

This helps management understand customer opinions about different products.

Write SQL queries to:

1. Display all reviews for a products.

```
SQL> SELECT * FROM Review WHERE ProductID = 'P302';

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0002        102 P302             4
Good football                                       21-JAN-26
```

2. Display customer name with their reviews.

```
SQL> SELECT Customer_Name, R.ReviewID, R.ProductID, R.Rating, R.ReviewTest FROM Customer C JOIN Review R ON C.Customer_ID = R.CustomerID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
REVIEWI PRODUCT     RATING REVIEWTEST
------- ------- ---------- --------------------------------------------------

AJAY
REV0001 P301            5 Excellent backpack

kavya
REV0002 P302            4 Good football

MANOJ
REV0003 P303            5 Excellent car cover


CUSTOMER_NAME
--------------------------------------------------------------------------------
REVIEWI PRODUCT     RATING REVIEWTEST
------- ------- ---------- --------------------------------------------------

NISHA
REV0004 P304            3 Good quality but could be improved
```

3. Find products having maximum reviews.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, COUNT(R.REVIEWID) AS REVIEW_COUNT FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PR
ODUCT_ID, P.PRODUCT_NAME ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME        REVIEW_COUNT
------- ------------------- ------------
P302    Room 102                       1
P304    Room 202                       1
P303    Room 201                       1
P301    Room 101                       1
```

4. Display recent customer feedback.

```
SQL> SELECT C. CUSTOMER_NAME, P.PRODUCT_NAME, R.RATING, R.REVIEWTEST AS CUSTOMER_FEEDBACK, R. REVIEWDATE FROM REVIEW R JOIN CUSTOMER C ON R.CUSTOMER
ID = C.CUSTOMER_ID JOIN PRODUCT P ON R.PRODUCTID = P.PRODUCT_ID ORDER BY R.REVIEWDATE DESC;

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME            RATING
------------------- ----------
CUSTOMER_FEEDBACK                                  REVIEWDAT
-------------------------------------------------- ---------
NISHA
Room 202                     3
Good quality but could be improved                 03-FEB-26

MANOJ
Room 201                     5
Excellent car cover                                02-FEB-26
```

5. Retrieve reviews with ratings above 4.

```
SQL> SELECT * FROM Review WHERE Rating > 4;

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0001        101 P301             5
Excellent backpack                                  20-JAN-26

REV0003        103 P303             5
Excellent car cover                                 02-FEB-26
```

Example Query:

6. Calculate Average Product Ratings

Aggregate functions are used to analyse product ratings. The average rating for each product is calculated using the AVG() function. The COUNT() function is used to find the number of reviews for each product. The system also identifies the highest-rated products and products with an average rating above 4.

Tasks:

1. Calculate average rating for each product.

```
SQL> SELECT P.PRODUCT_NAME, AVG(R.RATING) AS AVERAGE_RATING FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME         AVERAGE_RATING
-------------------- --------------
Room 101                          5
Room 202                          3
Room 102                          4
Room 201                          5
```

2. Count the number of reviews per product.

```
SQL> SELECT PRODUCTID, AVG(RATING) AS AVERAGE_RATING FROM REVIEW GROUP BY PRODUCTID HAVING AVG(RATING) > 4;

PRODUCT AVERAGE_RATING
------- --------------
P303                 5
P301                 5
```

3.Identify products with average rating above 4.

```
SQL> SELECT PRODUCTID,AVG(RATING) AS AVERAGE_RATING FROM Review GROUP BY
 PRODUCTID ORDER BY AVERAGE_RATING DESC;

PRODUCT AVERAGE_RATING
------- --------------
P303                 5
P301                 5
P302                 4
P304                 3
```

7. Identify Highly Rated Products

Product rating reports are generated to identify products that receive positive feedback from customers. These reports help compare products based on the number of reviews and average ratings. Highly rated products can be identified easily through the analysis.

Report 1: Product Rating Analysis Display:

- Product name
- Number of reviews
- Average rating

```
SQL> SELECT P.PRODUCT_NAME,COUNT(R.REVIEWID) AS NUMBER_OF_REVIEWS,ROUND(
AVG(R.RATING),2) AS AVERAGE_RATING FROM Product P LEFT JOIN Review R ON
P.PRODUCT_ID=R.PRODUCTID GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME         NUMBER_OF_REVIEWS AVERAGE_RATING
-------------------- ----------------- --------------
Room 302                             0
ROOM 61                              0
Room 402                             0
Room 101                             1              5
Room 202                             1              3
Room 102                             1              4
Room 201                             1              5
Room 501                             0
Room 601                             0
Room 301                             0

10 rows selected.
```

Report 2: Customer Feedback Analysis

Find:

- Most reviewed products.

```
SQL> SELECT P.PRODUCT_NAME,COUNT(R.REVIEWID) AS NUMBER_OF_REVIEWS FROM P
roduct P JOIN Review R ON P.PRODUCT_ID=R.PRODUCTID GROUP BY P.PRODUCT_NA
ME ORDER BY NUMBER_OF_REVIEWS DESC;

PRODUCT_NAME         NUMBER_OF_REVIEWS
-------------------- -----------------
Room 101                             1
Room 201                             1
Room 102                             1
Room 202                             1
```

- Highly rated products

```
SQL> SELECT P.PRODUCT_NAME,ROUND(AVG(R.RATING),2) AS AVERAGE_RATING FROM
 Product P JOIN Review R ON P.PRODUCT_ID=R.PRODUCTID GROUP BY P.PRODUCT_
NAME HAVING AVG(R.RATING)>=4;

PRODUCT_NAME         AVERAGE_RATING
-------------------- --------------
Room 101                          5
Room 102                          4
Room 201                          5
```

Report 3: Rating Distribution

- Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS FROM Review WHERE RATING=5;

FIVE_STAR_RATINGS
-----------------
                2
```

- Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS FROM Review WHERE RATING=4;

FOUR_STAR_RATINGS
-----------------
                1
```

- Number of low-rated products

```
SQL> SELECT COUNT(*) AS LOW_RATED_REVIEWS FROM Review WHERE RATING<=2;

LOW_RATED_REVIEWS
-----------------
                0
```

- Complete Rating

```
SQL> SELECT RATING,COUNT(*) AS NUMBER_OF_RATINGS FROM Review GROUP BY RA
TING ORDER BY RATING DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 2
         4                 1
         3                 1
```

9. Result

The Product Review and Rating Management System was implemented using SQL queries. Customer reviews and ratings were stored, updated, retrieved, and analysed successfully. Product rating reports were also generated.

10. Conclusion

Thus, the Product Review and Rating Management System was successfully implemented. It helps collect customer feedback, analyse product ratings, identify highly rated products, and find products that require improvement. This system supports better product quality and customer satisfaction.